



Aspen HYSYS® Certified User Exam

* Obligatorio

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6

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Exam Instructions

This exam is divided into two sections: Section 1 measures your understanding of the concepts and Section 2 consists of a lab. The exam is multiple choice with only one correct answer per question. You will have up to 4 hours to complete the exam. Aids are permitted during the exam and limited to AspenTech training material and AspenTech software. Mobile phones must be shut off and put away. Good luck!

7

If a user adds a component list using HYSYS Databank, which Property Package database/s can be used to define the Fluid Package? (1 Punto)

- ☒ A. HYSYS Databank
- ☐ B. Only Aspen Properties
- ☐ C. COMThermo
- ☐ D. Either A or C

8

For Aspen HYSYS to estimate the hypothetical component’s unknown properties, the following minimum information must be supplied (1 Punto)

- ☐ A. Enthalpy
- ☐ B. Molecular Weight and Liquid Density
- ☐ C. Normal Boiling Pt
- ☒ D. Either B or C

9

What type of information is exported as a binary file with the extension ‘.fpk’? (1 Punto)

- ☐ A. Component list defined in the Component Lists folder only
- ☐ B. Property Package selected including binary coefficients only
- ☒ C. Component list defined and Property Method Package Selected
- ☐ D. None of the above

10

What kind of options can be used for characterizing a crude assay with TBP distillation data in Aspen HYSYS? (1 Punto)

- ☐ A. Correlation Manager

- ☒ B. HYSYS Petroleum Assay Manager
- ☐ C. Component list with hypothetical component
- ☐ D. Either A or B

11

What are the necessary steps to characterize a crude assay using Aspen Petroleum Assay Manager? (1 Punto)

- ☒ A. Characterize the assay, specify/import assay data, and attach assay data with a material stream
- ☐ B. Specify/import assay data, characterize the assay, and attach assay data with a material stream
- ☐ C. Install material stream and specify/import assay data
- ☐ D. Import assay data and attach assay data with a material stream

12

What is the main difference between the Oil Manager and Petroleum Assay Management characterization tools? (1 Punto)

- ☐ A. Both Oil Manager and Assay Manager uses the cutpoint on Midpoint basis
- ☐ B. Oil Manager uses end point basis
- ☐ C. Assay Manager uses End Point basis
- ☐ D. Oil Manager does not create hypo components for its blend

13

What is the minimum information needed to solve a material stream? (1 Punto)

- ☐ A. Composition and temperature
- ☐ B. Mass flowrate and temperature
- ☒ C. Composition, temperature, pressure, and flowrate (mass/molar/volumetric)
- ☐ D. Composition and flowrate (mass/molar/volumetric)

14

Identify the mathematical/logical operations in Aspen HYSYS from the following: (1 Punto)

- ☐ A. Pump, Cooler
- ☒ B. Balance, Adjust, Set
- ☐ C. Balance, Heater
- ☐ D. Set, User Property

15

A user needs to achieve a target liquid product rate from the separator by manipulating the feed temperature. Select the logical operation(s) that should be used. (1 Punto)

- ☐ A. Recycle
- ☐ B. Set
- ☒ C. Adjust
- ☐ D. B and C

16

A Recycle operation should be used when (1 Punto)

- ☒ A. Downstream material streams are mixed with upstream material streams
- ☐ B. Closed loop flowsheet
- ☐ C. Both A & B
- ☐ D. Either A or B

17

Which of the following is TRUE regarding Adjust Operation? (1 Punto)

- ☐ A. The Independent variable cannot be a calculated value; it must be specified.
- ☐ B. The Independent variable can be a calculated value.
- ☐ C. The Adjust is a steady state operation; HYSYS ignores it in dynamic mode.
- ☒ D. Both A&C

18

Identify the types of Stream Analysis in Aspen HYSYS types from the following: (1 Punto)

- ☒ A. Hydrate Formation, Envelope, Critical Properties
- ☐ B. User Property, Correlation Manager, Oil Manager
- ☐ C. Either A or B
- ☐ D. Both A & B

19

If a user wants to review the trend of the stream density along with the stream T & P, how can that be achieved in Aspen HYSYS? (1 Punto)

- ☐ A. Open the stream, add the stream analysis from Attachments | Analysis, then select Property Table.
- ☒ B. Add the Stream Analysis - Property Table to this stream from the Home Ribbon.

- ☐ C. Add a Property Table on the Stream Analysis folder in the left Navigation.
- ☐ D. Either A, B or C

20

If a user has petroleum assay attached to a stream and would like to check the ASTM D86 boiling curve and TBP of the petroleum product stream, which of the following tools should be used? (1 Punto)

- ☐ A. Stream Analysis tools - Boiling Point Curves
- ☐ B. Stream Analysis tools - Petroleum Assay
- ☐ C. Stream Analysis tools - Property Table
- ☐ D. Case Studies

21

Which of the following statements is FALSE regarding Envelope Stream Analysis? (1 Punto)

- ☐ A. Using Envelope Analysis, Vapor-Liquid Envelopes can be plotted for Pressure-Enthalpy, Pressure-Temperature variables.
- ☐ B. The Envelope Analysis is restricted to the Peng-Robinson, SRK, and Glycol property packages.
- ☐ C. A hydrate curve can be plotted with the Envelope Analysis.
- ☐ D. The Envelope Analysis is restricted to the Peng-Robinson, SRK, SourPR, and SourSRK.

22

While building an open loop flowsheet, the user decides to mix a downstream material stream with the upstream material. The user can include the Recycle operation in the flowsheet by (1 Punto)

Procedure (1) Estimate the recycle stream, then solve the flowsheet, then add a recycle operation and finally attach the estimated and calculated streams
Procedure (2) Add a recycle operation, then attach the calculated and estimated streams and finally connect the estimated stream to the flowsheet

- ☐ A. Procedure (1) ALONE is sufficient, but Procedure (2) alone is NOT sufficient.
- ☐ B. Procedure (2) ALONE is sufficient, but Procedure (1) alone is NOT sufficient.
- ☐ C. BOTH Procedure TOGETHER are sufficient, but NEITHER Procedure ALONE is sufficient.
- ☐ D. EACH Procedure ALONE is sufficient.

23

If a user has multiple feeds with one vapor and two liquid product streams, which of the following separator operations should be selected? (1 Punto)

- ☐ A. Separator
- ☐ B. Tank

☐ C. 3-Phase Separator

☐ D. 2-Phase Separator

24

Which of the following simulation analysis tool allows the user to monitor the steady state response of key variables to changes in their process? (1 Punto)

☐ A. Property Table

☒ B. Case Study

☐ C. Workbook

☐ D. None of the above

25

If a user has setup a Nested Case Study for 2 independent variables (Variable A & B) to monitor the response of 1 dependent variable and the number of steps for Variable A is 7 and for Variable B is 4. How many total states will the case study run for? (1 Punto)

☐ 11

☒ 28

☐ 22

☐ 32

26

What type of exchanger specification should be provided if the user wants to include the minimum temperature difference between the hot and cold stream (not necessarily at the inlet or outlet)? (1 Punto)

☐ A. Delta Temperature

☐ B. Minimum Approach

☒ C. LMTD

☐ D. Duty

27

A simple heat exchanger unit is required in the simulation to estimate the UA for a gas condensing process. No sizing data is available for the unit. What type of heat exchanger model is required? (1 Punto)

☒ A. Simple Weighted

☐ B. Simple End Point

☐ C. Simple Steady State Rating

☐ D. Either B or C

28

There are several correlation methods for calculating pressure drop of a pipe segment, which of the following methods could be used for flow through a horizontal pipe with expected liquid hold-up calculated and the ability to define a flow regime map? (1 Punto)

- ☐ A. Beggs & Brill (1973 or 1979)
- ☐ B. HTFS Homogeneous, HTFS Liquid Slip
- ☐ C. Aziz, Govier, & Fogarasi
- ☐ D. Either A or C

29

A pipe is buried 1.5 m deep underground and has a Urethane Foam insulation of 0.05 m, which of the following heat transfer options can be selected for performing the rigorous heat transfer calculation for this pipe operation? (1 Punto)

- ☐ A. Overall HTC option and Heat Loss option to calculate heat transfer
- ☐ B. Segment HTC option to define heat transfer for each pipe segment
- ☒ C. Estimate HTC option
- ☐ D. None of the above

30

Which of the following is correct for a 100 m long horizontal pipe segment? (1 Punto)

- ☐ A. Elevation is -100.0
- ☐ B. Elevation is 100.0
- ☒ C. Elevation is 0.0
- ☐ D. None of the above

31

If the fluid temperature is lower than the hydrate formation temperature, then there is a potential problem in the process. How does one check the hydrate formation temperature in the pipeline? (1 Punto)

- ☒ A. Use the Hydrate Formation Stream Analysis tool for the inlet and out streams
- ☒ B. Perform Hydrates Profile Calculation (Flow Assurance) in the pipe
- ☐ C. Perform Slug Analysis or Wax Deposition
- ☒ D. Either A or B

32

Which of the following Pipe Flow Assurance studies can be applied for studying the continuous impact of particles (liquids or solids) on the pipe wall surface? (1 Punto)

- ☐ A. CO2 Corrosion
- ☒ B. Wax Deposition
- ☐ C. Erosion
- ☐ D. Either A or B

33

A compressor operation has curves at 4 different speeds. The inlet and outlet stream compositions, inlet temperature, and inlet flowrate are known. What additional inputs are required to solve the compressor? (1 Punto)

- ☐ A. Discharge pressure
- ☐ B. Speed
- ☐ C. Discharge temperature
- ☒ D. Either A or B

34

In Aspen HYSYS, a heat exchanger block has been added onto the flowsheet and inlet/outlet streams have been connected. Both the hot side and cold side inlet streams have been fully solved. Meanwhile, we have provided the pressure drop on the shell side and tube side of the exchanger. What else do we need to specify in order to solve the heat exchanger block? (1 Punto)

Statement (1) The duty of the exchanger. ✓
Statement (2) The outlet temperature of shell side stream.

- ☐ A. Statement (1) ALONE is sufficient, but statement (2) alone is not sufficient.
- ☐ B. Statement (2) ALONE is sufficient, but statement (1) alone is not sufficient.
- ☐ C. BOTH statements TOGETHER are sufficient, but NEITHER statement ALONE is sufficient.
- ☒ D. EACH statement ALONE is sufficient. ✓
- ☐ E. Statements (1) and (2) TOGETHER are NOT sufficient.

35

The Reboiled Absorber template consists of _____. (1 Punto)

- ☒ A. a Tower and a bottom reboiler.
- ☐ B. a Tower and an overhead Condenser (partial or total).
- ☐ C. Both A & B

☐ D. Either A or B

36

The degrees of freedom for a refluxed column with a partial condenser and a liquid pumparound from stage 6 to stage 5 is_____. (1 Punto)

☐ 1

☐ 2

☐ 3

☒ 4

37

Which of the following Column Specification Types should be used to specify the molar ratio of a set of components over another set of components for the liquid phase on any stage? (1 Punto)

☐ A. Component Recovery

☐ B. Component Ratio

☒ C. Component Fractions

☐ D. Component Flow Rate

38

Which of the following reasons apply here for the column unconverged status due of Degrees of Freedom as shown below? (1 Punto)

☐ A. Column is unconverged as fewer column specifications are given than needed

☐ B. Column is unconverged as more column specifications are given than needed

☐ C. Column is unconverged as no column specifications are given

☐ D. None of the above

39

In order to view flowrate in tonnes/hr, what are the necessary steps to be taken? (1 Punto)

☐ A. Go to File | Options and select SI Units of Measure default set

☐ B. Go to File | Options, copy SI Units of Measure default set to create custom set, change the units of Mass Flow to tonnes/hr

☐ C. Go to File | Options and select Field Units of Measure default set

☐ D. Go to File | Options, copy Field Units of Measure default set to create custom set, change the units of Molar Flow to tonnes/hr

40

Which of the following is TRUE about the Spreadsheet Operation? (1 Punto)

- ☒ A. Drag & drop variable values from a stream or unit operation to the Spreadsheet
- ☐ B. Only variables that are calculated by the Spreadsheet are exportable to the flowsheet
- ☐ C. A variety of mathematical operations are supported, such as formulas, functions, inequalities, logical operations, IF/THEN/ELSE statements, etc.
- ☐ D. All of the above

41

How can a user customize General user preferences, options, and default settings in Aspen HYSYS? (1 Punto)

- ☐ A. The user can customize/configure the general options and default setting under File | Options.
- ☐ B. The user can customize General user preferences from the Customize Ribbon.
- ☐ C. The user can customize General user preferences and default settings in each Unit Operation and Stream.
- ☐ D. The user cannot configure the general options and default settings.

42

If the inlet stream of a cooler is fully specified, what parameters should be defined to solve the cooler? (1 Punto)

- ☐ A. Delta P, Delta T, and Duty
- ☐ B. Duty and outlet stream T
- ☒ C. Delta P and Delta T
- ☐ D. Outlet stream pressure and Delta P

43

There is a V-L Vessel, inlet stream pressure is 6156.5 kPa, Vapour outlet stream pressure is 6106.5 kPa and liquid outlet stream pressure is 6136.5 kPa, how can a user define this vessel Delta P to obtain the correct pressure profile? (1 Punto)

- ☐ A. Define Delta P for Inlet as 50 kPa and Delta P for Vapour outlet as 0 kPa.
- ☐ B. Define Delta P for Inlet as 20 kPa and Delta P for Vapour outlet as 30 kPa.
- ☐ C. Define Delta P for Inlet as 30 kPa and Delta P for Vapour outlet as 20 kPa.
- ☐ D. Define Delta P for Inlet as 0 kPa and Delta P for Vapour outlet as 50 kPa.

44

Cooling water is used to cool a gas stream. All heat exchanger inlet streams are fully specified, both shell and tube side pressure drops are given, and gas stream outlet T is specified. The

heat exchanger is converged, however, after checking the performance, the user found that Min. Approach is 30°C, and the duty is lower than expected. Which of the following is the best option to maximize the heat exchanger duty? (1 Punto)

- ☐ A. Delete the gas stream outlet T, define LMTD as the Specification to effectively use the cold utility.
- ☒ B. Adjust the gas stream outlet T until the Min. Approach is lower to effectively use the cold utility.
- ☐ C. Delete the gas stream outlet T, define Min. Approach larger than 5°C but smaller than 30°C as the Specification to effectively use the cold utility.
- ☐ D. There is no way to optimize the heat exchanger because all parameters are defined.

45

Which kind of heat transfer operations can be integrated with Aspen Exchanger Design and Rating (EDR) tools? (1 Punto)

- ☐ A. Heat Exchanger, Cooler, Heater
- ☐ B. Heat Exchanger, Fired Heater
- ☐ C. Air Cooler, LNG Exchanger
- ☒ D. Both B and C

46

What kind of report can be added to the flowsheet? (1 Punto)

- ☒ A. Data Table
- ☐ B. Workbook
- ☐ C. Model Summary
- ☐ D. Both A & B

47

Which of the following can be sources for Sub-Flowsheet? (1 Punto)

- ☐ A. Previously saved process template file (*.tpl)
- ☐ B. Previously saved HYSYS flowsheet file (*.hfl)
- ☐ C. Previously saved HYSYS compound file (*.hscz)
- ☐ D. Both A & B

48

What does the below message indicate? (1 Punto)

Consistency Error

A consistency error has occurred

A consistency error has occurred. This can arise either when two objects calculate differing values for the same variable or one object's calculations are conflicting with existing specifications.

In general, this means that the simulation is overspecified.

However, if the discrepancy is small and one of the values is calculated by a column flowsheet, tightening the column solver tolerances may eliminate the inconsistency.

Variable Information

Name of Inconsistent Variable	Pressure
Name of Object	Process In
Object Type	Material Stream

Calculation Source Information

Old Calculation

2100 kPa
Specified

Value	
Calculated By Object Name	
Calculated By Object Type	
Specification / Calculation	

New Calculation

2140 kPa
VLV-101
Valve
Calculated

☐ A. Stream 'Process In' Pressure is underspecified

☐ B. Stream 'Process In' Pressure is overspecified

☐ C. Solver is 'On Hold'

☒ D. Both B & C



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Section 2: Aspen HYSYS Lab

Problem Statement:

You have been tasked to construct a simulation for the treatment of an NGL feed. The NGL will be processed in a depropanizer column for the recovery of desired hydrocarbon products. The feed contains light gases which must be separated before feeding the depropanizer column.

Process Description:

NGL is first separated from vapor in a 2-phase separator. The light gases from the separator will be compressed before exporting to the pipeline. The liquid product from the separator is pumped to a heat exchanger and then to a heater before feeding the depropanizer column. The column will separate most light gases (ethane and propane). The bottom liquid product will be transported to a third-party facility for further fractionation. You will not be simulating the third-party facility.

The depropanizer has 24 stages, a reboiler, a partial condenser, and has an operating pressure of 18.25 bar_g. The specification of the column is adjusted such that the top product contains mostly light products (ethane and propane).

The necessary input for building the flowsheet is provided in the Input Data section.

To complete the lab, you will start with a new case, add the components, select the correct fluid package, and build the flowsheet according to the Flowsheet image and Input Data provided in the next section. After building the process flowsheet, answer questions 52-67 that follow.

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VLV-100	Control Valve
Delta P [kPa]	70
Separator	Separator
Delta P – Inlet [kPa]	0
Delta P – Vapour Outlet [kPa]	0
Feed Pump	Pump
Delta P [kPa]	1900
VLV-101	Control Valve
Delta P [kPa]	70
Heat Exchanger	Heat Exchanger
Heat Exchanger Model	Simple Weighted
Shell Side Delta P [kPa]	70
Tube Side Delta P [kPa]	70
Shell Out T [°C]	121.2

☐ yes, I've entered the above unit operation data

51

Unit Operation Data

☐ yes, I've entered the above unit operation data

52

What is the duty and LMTD of the Heat Exchanger? (4 puntos)

- ☐ A. Duty less than 4.920e5 kJ/h and LMTD less than 100C.
- ☐ B. Duty more than 4.920e6 kJ/h and LMTD less than 100C.
- ☐ C. Duty between 4.920e6 kJ/h and 5.20e6 kJ/h and LMTD between 120C and 125C
- ☐ D. Duty between 4.920e5 kJ/h and 5.20e5 kJ/h and LMTD between 100C and 110C

53

If there is requirement to restrict the LMTD of the Heat Exchanger to 118°C, what is the duty of the Heat Exchanger? (4 puntos)

- ☐ A. Duty less than 4.920e5 kJ/h
- ☐ B. Duty more than 5.20e6 kJ/h
- ☐ C. Duty more than 4.920e7 kJ/h
- ☐ D. Duty between 4.920e5 kJ/h and 5.20e5 kJ/h

54

The Cricondentherm and Two-Phase Critical Temperature for the NGL Feed stream are: (3 puntos)

- ☐ A. Cricondentherm is above 190°C and Two-Phase Critical Temperature is below 190°C
- ☐ B. Cricondentherm is above 170°C and Two-Phase Critical Temperature is above 170°C
- ☐ C. Cricondentherm is below 170°C and Two-Phase Critical Temperature is below 170°C
- ☐ D. None of the above

55

What is the mole fraction of C2 +C3 in Propane stream? (4 puntos)

- ☐ A. Less than 0.2
- ☐ B. Between 0.2 and 0.6
- ☐ C. Between 0.85 and 0.95
- ☐ D. More than 0.95

56

What is the Reboiler and Condenser duty of the Depropanizer? (4 puntos)

- ☐ A. Condenser Duty less than 2.10e6 kJ/h and Reboiler duty more than 5.70e7 kJ/h.
- ☐ B. Condenser Duty more than 2.10e8 kJ/h and Reboiler duty less than 5.70e8 kJ/h.
- ☐ C. Condenser Duty between 2.10e7 kJ/h and 3.101e7 kJ/h and Reboiler duty between 4.70e7 kJ/h and 4.90e7 kJ/h
- ☐ D. Condenser Duty between 2.00e7 kJ/h and 2.25e7 kJ/h and Reboiler duty between 5.60e7 kJ/h and 5.90e7 kJ/h

57

You are required to modify the process so that the lights from the top of the column contains mostly ethane and propane. Add a new specification such that the mole fraction for C2+C3 in the Propane stream is 0.98. The reflux ratio is still mainlined at 2. How does the flowrate of stream C4+ change? (3 puntos)

- ☐ A. Flowrate did not change
- ☐ B. Flowrate increased
- ☐ C. Flowrate decreased

58

How do the Reboiler and Condenser duties change after the new specification modification? (3 puntos)

- ☐ A. Reboiler and Condenser duty did not change
- ☐ B. Reboiler duty increased and Condenser duty decreased
- ☐ C. Reboiler duty decreased and Condenser duty decreased

☐ D. Reboiler duty increased and Condenser duty increased

59

The condenser and reboiler use utilities for cooling and heating. Report the cooling water flow rate in the condenser, and HP steam flowrate in the reboiler. (4 puntos)



- ☐ A. Cooling Water Flowrate is more than 9.10e5 kg/h and HP Steam Flowrate is less than 3.10e3 kg/h
- ☐ B. Cooling Water Flowrate is less than 9.10e4 kg/h and HP Steam Flowrate is more than 3.17e5 kg/h
- ☐ C. Cooling Water Flowrate is more than 9.10e5 kg/h and HP Steam Flowrate is more than 3.00e5 kg/h
- ☐ D. Cooling Water Flowrate is less than 9.10e5 kg/h and HP Steam Flowrate is more than 3.00e4 kg/h

60

What is the power of the compressor and pump? (4 puntos)



- ☐ A. Compressor Power is less than 40 kW and Pump Power is less than 130 kW
- ☐ B. Compressor Power is more than 40 kW and Pump Power is less than 130 kW
- ☐ C. Compressor Power is less than 45 kW and Pump Power is less than 100 kW
- ☐ D. Compressor Power is more than 40 kW and Pump Power is more than 130 kW

61

You have just received the compressor curves from the vendor (see table below). You will add the curves for accurate calculations of efficiency, power and speed of the compressor. Report the compressor power, speed, and adiabatic efficiency after implementing the curves. How do the curves impact the Adiabatic efficiency of the Compressor? (3 puntos)



- ☐ A. Compressor Efficiency decreased
- ☐ B. Compressor Efficiency increased
- ☐ C. Compressor Efficiency had no impact

62

You are required to calculate the dewpoint temperature of the Process Out stream using a Balance Operation at the same pressure, flow rate and composition. What is the calculated dew point temperature? (4 puntos)



- ☐ A. Less than 135C
- ☐ B. Between 125C and 135C
- ☐ C. Between 135C and 140C
- ☐ D. More than 140C

63

You are required to present the overall mass, energy balance and utilities (cooling water and steam) consumption for the flowsheet. How can this be presented without doing any manual calculations? (3 puntos)

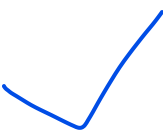
- ☐ A. Using Model Summary
- ☐ B. Using Flowsheet Summary
- ☐ C. Both A & B
- ☐ D. None of the above



64

Note the Overall Heat loss of the Pipe -100. If insulation (Urethane Foam) of 0.05 m thickness is used while the pipe is still buried 1 m deep in Ground, how does the heat loss of the pipe change? (3 puntos)

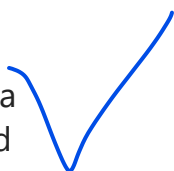
- ☐ A. Heat Loss increased
- ☐ B. Heat Loss decreased
- ☐ C. Heat Loss does not change



65

You are required to study the impact of NGL feed throughput to heat exchanger, pump and compressor. Configure a case study for NGL feed flow rate from 160 m3/h to 280 m3/h with a step of 20 m3/h and report the power requirements by the compressor and pump. If the feed flow rate is 250 m3/h, then the pump power is _____. (4 puntos)

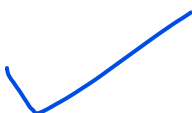
- ☐ A. More than 200 KW
- ☐ B. Between 120 and 140 KW
- ☐ C. Between 155 and 170 KW
- ☐ D. Less than 120 KW



66

Using the Rigorous Shell & Tube exchanger model, design the exchanger (E-100) for the following specifications and determine the duty of the Heat Exchanger. (4 puntos)

- ☐ A. Duty less than 4.50e5 kJ/h
- ☐ B. Duty between 4.50e5 kJ/h and 5.50e5 kJ/h
- ☐ C. Duty more than 5.20e7 kJ/h
- ☐ D. Duty between 5.20e6 kJ/h and 5.50e6 kJ/h



67

If the COL Feed temperature is targeted to be 55C, what should be the NGL feed flowrate?
(4 puntos)



- ☐ A. Flowrate less than 150 m3/h
- ☐ B. Flowrate more than 180 m3/h
- ☐ C. Flowrate is between 150 m3/h and 180 m3/h
- ☐ D. None of the above

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